Hugo Fruchet

PhD student in machine learning | ISIR (Sorbonne Université) & INRIA (ENS-PSL), Paris

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Education

PhD in Machine Learning **2026 – present**
Sorbonne Université, ISIR & INRIA ARGO team Paris, France
- Topic: theoretical foundations of scalable reinforcement learning and LLM post-training
- Advisors: Edouard Oyallon and Marc Lelarge

École Normale Supérieure Paris-Saclay **2022 – 2026**
Diplôme de l'ENS, with courses at ENS Ulm, Polytechnique & Collège de France Gif-sur-Yvette, France
- **ARTeQ**: research year in quantum computing and quantum machine learning
- **M2 MVA** (Mathematics, Vision, Learning), research track: deep learning theory, probabilistic models, optimal transport, RL, bandits & online learning
- **L3 + M1 MPRI**: statistical learning, deep learning, computer vision; parallel programming (CUDA), theoretical CS (semantics, λ-calculus, logic, computability)

Preparatory classes MPSI/MP* **2019 – 2022**
Intensive two-year program in mathematics & computer science Bordeaux, France

Research experience

Pre-doctoral internship in machine learning **2026/02 – 2026/09**
Understanding and improving large reasoning models ISIR, Paris, France
- Developed interpretability methods for LLM reasoning; worked on improving reasoning efficiency

M2 internship in hybrid quantum-classical ML **2025/04 – 2025/08**
Designing a photonic quantum convolutional neural network Quandela, Massy, France
- Proposed a novel convolutional layer; studied trainability issues (vanishing gradients / barren plateaus)
- Results presented at QTML 2025 (poster) and Quantum AI 2026 (talk, presented by a co-author).

M1 internship in machine learning **2024/04 – 2024/07**
Bayesian inference under differential privacy Université de Neuchâtel, Switzerland

Teaching

Teaching assistant **2026 – 2027**
Sorbonne Université Paris, France
- **Data Structures** (L2, *Structures de données*): computer lab sessions
- **Deep Learning** (M2 MIND, DEEPL): practical sessions

Publications & presentations

D. Wang, A. Walsh, **H. Fruchet**, A. Soret, P.-E. Emeriau, and S. Mansfield, "Quantum Circular Convolutional Neural Networks (QCirCNN): A Parameter and Energy Efficient Model for Computer Vision," in *Quantum AI Conference 2026, contributed talk*, Indianapolis, IN, USA, 2026.

D. Wang, A. Walsh, **H. Fruchet**, A. Soret, and P.-E. Emeriau, "QCirCNN: a photonic-native quantum circular convolution network based on circulant matrices," in *Quantum Techniques in Machine Learning (QTML), poster session*, Singapore, 2025.

Projects

- **Truthfulness interpretability in transformers**: generalized latent-knowledge probing methods (Burns et al., 2022; Marks & Tegmark, 2023)
- **Coherence-aware diffusion**: improved and benchmarked the method of arXiv:2405.20324

Skills

Machine learning: PyTorch, scikit-learn
Programming: Python, C/C++, CUDA, OCaml, Rust
Tools: Linux, Git
Languages: French (native), English (C1), Spanish (beginner)

Other

Saxophonist and drummer, f[ENS]foire student brass band 2022 – 2026